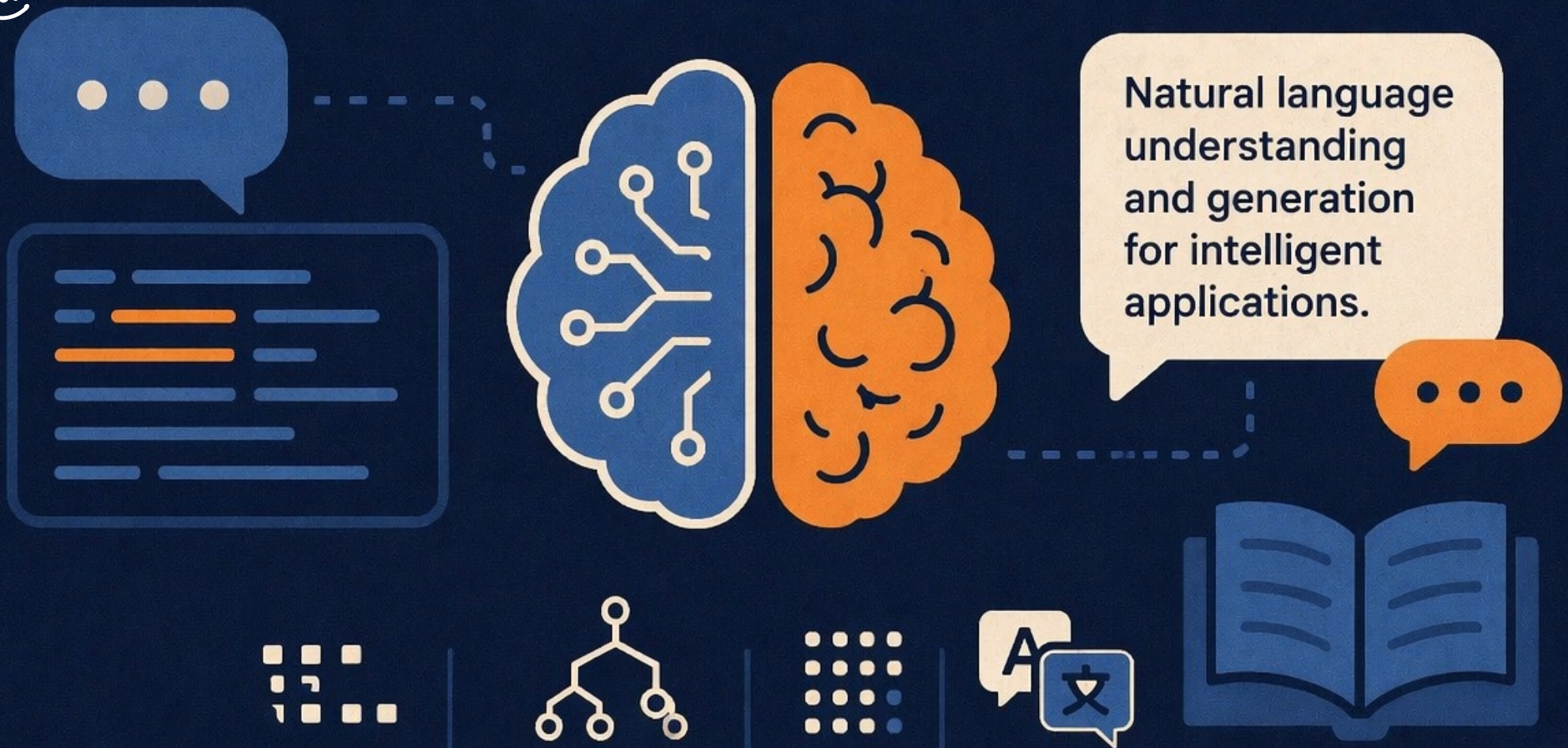
NATURAL LANGUAGE PROCESSING

class starts @ 9:05 pm



Lecture: RNN Continued

#Agenda

1) What is RNN?
2) Forward propagation
3) Backward propagation
4) Coding.

RNN

↳ Recurrent Neural Network

ANN — Tabular Data

CNN — grid structure image

RNN — Sequential Data

Timeseries data

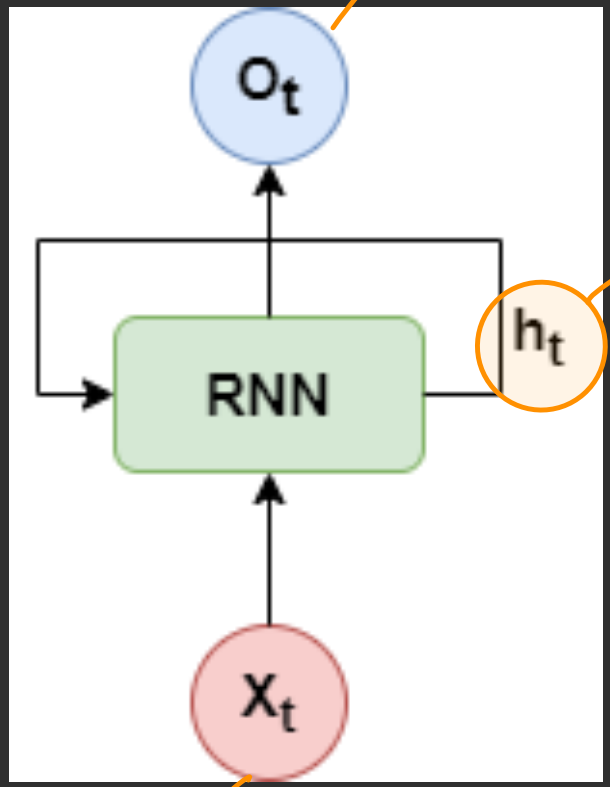
DNA pattern

Why we need RNN?

1) Text length are not same.

2) The idea of sequence is lost.

#RNN

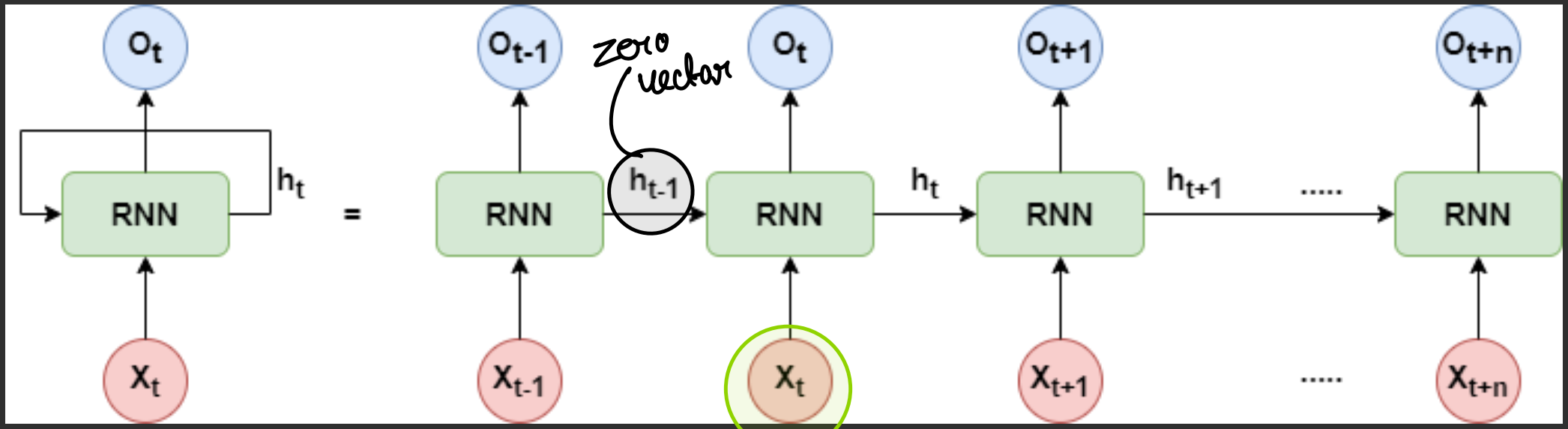


Output at time t

hidden state at time t

Input at time t

RNN unravelled over time

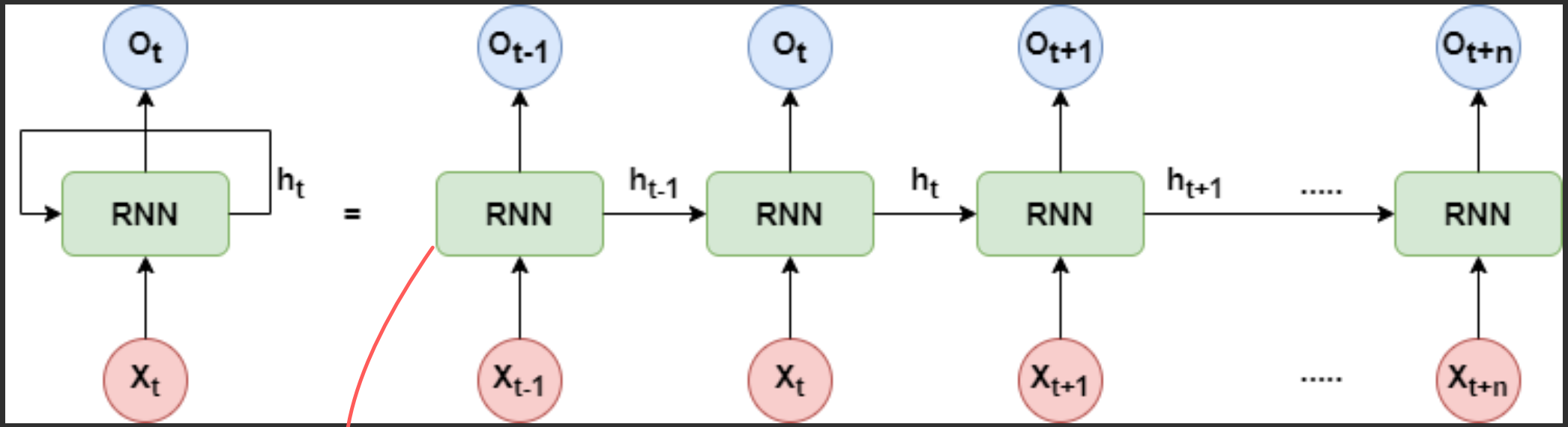


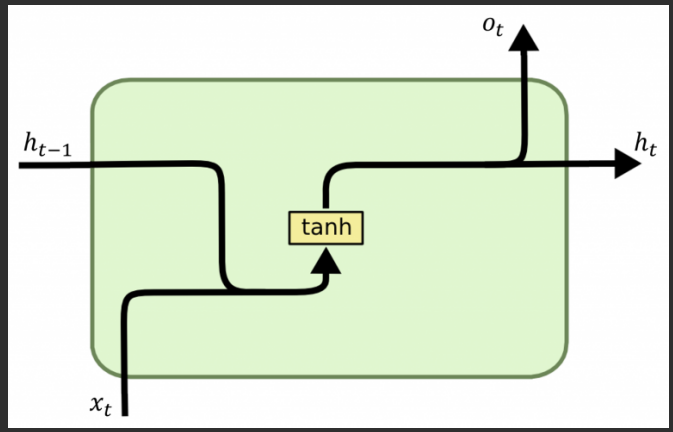
[1 0 0 1 0]

Hello, my day way good

→ encode / embedding (Word2Vec / OpenAI)

OHE


O_t
RNN
h_t
=
O_{t-1}
O_t
O_{t+1}
O_{t+n}
h_{t-1}
h_t
h_{t+1}
.....
RNN
X_t
X_{t-1}
X_t
X_{t+1}
X_{t+n}


o_t
h_{t-1}
h_t
tanh
x_t

any value ⟶ {-1, 1}

{0, 1} sigmoid

↳ no range

Vanishing gradient

↳ range based

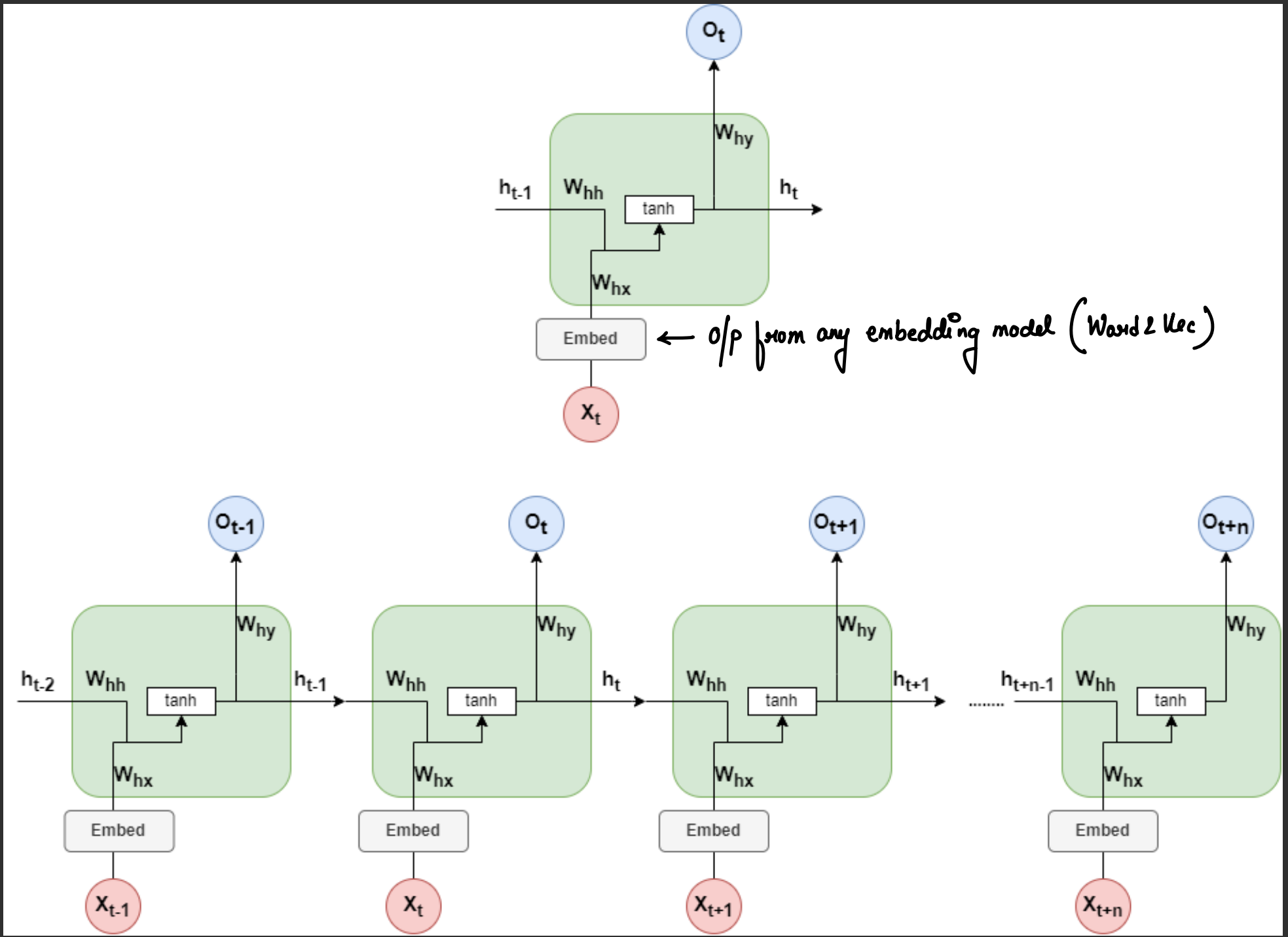
Spam Vs Ham

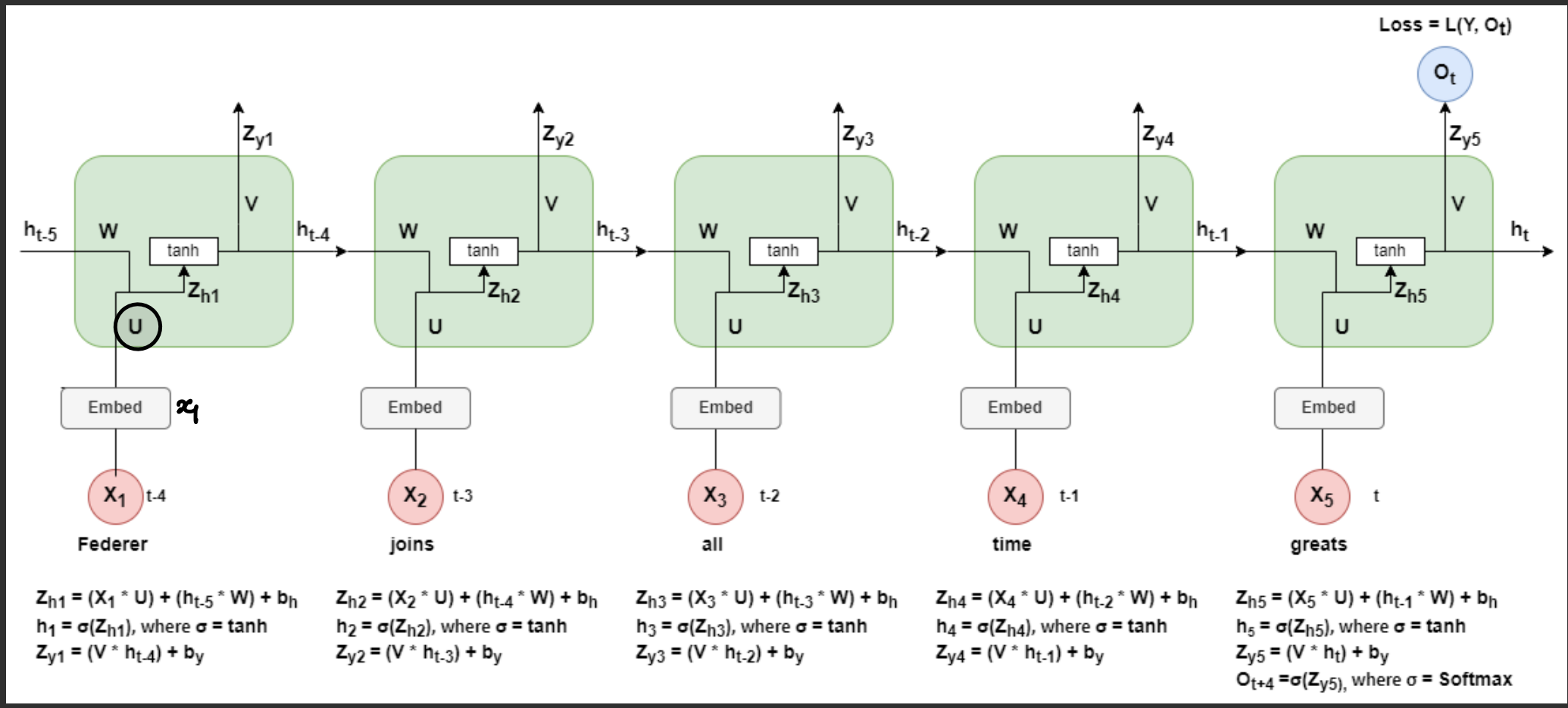
-1

Stole
need
loan
help

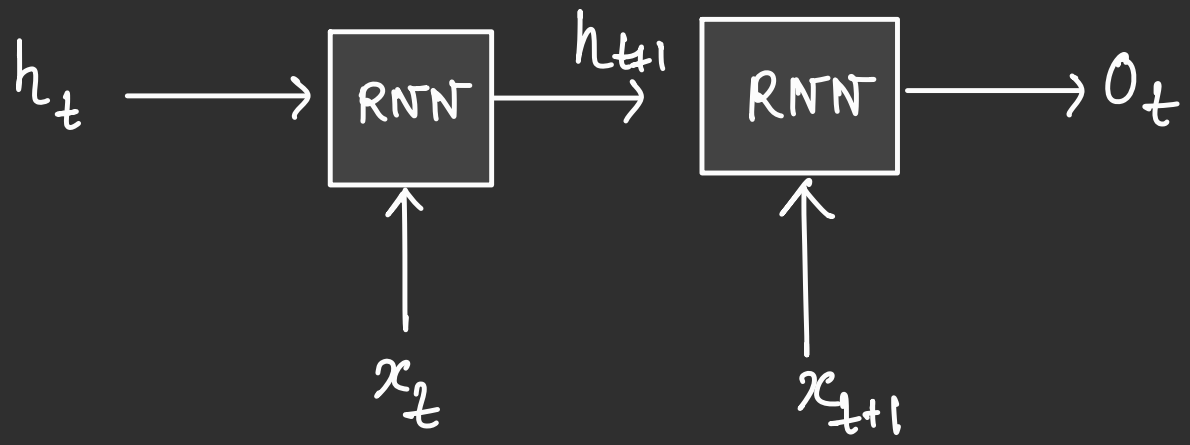
+1

email


O_t
W_{hy}
h_{t-1}
W_{hh}
tanh
h_t
W_{hx}
Embed
← o/p from any embedding model (Word2Vec)
X_t
O_{t-1}
O_t
O_{t+1}
O_{t+n}
h_{t-2}
h_{t-1}
h_t
h_{t+1}
h_{t+n-1}
.....
X_{t-1}
X_t
X_{t+1}
X_{t+n}

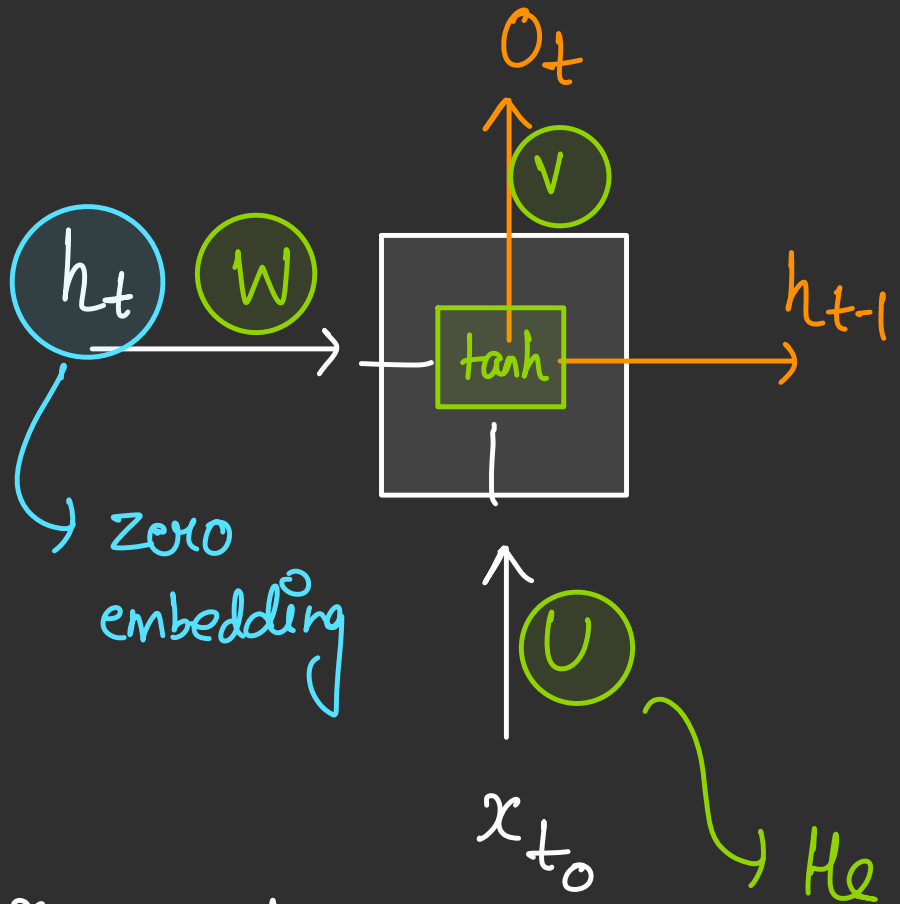


$$\begin{cases} W_{hx} = U \\ W_{hy} = V \\ W_{hh} = W \end{cases}$$


h_t
RNN
h_{t+1}
RNN
o_t
x_t
x_{t+1}

R_1 cat mat rat ⟶ [1 0 0] [0 1 0] [0 0 1]

R_2 mat rat mat ⟶ [0 1 0] [0 0 1] [0 1 0]



$$z_1 = (x_t \cdot U) + (h_t \cdot W) + b_1$$

$$h_t = \tanh(z_1)$$

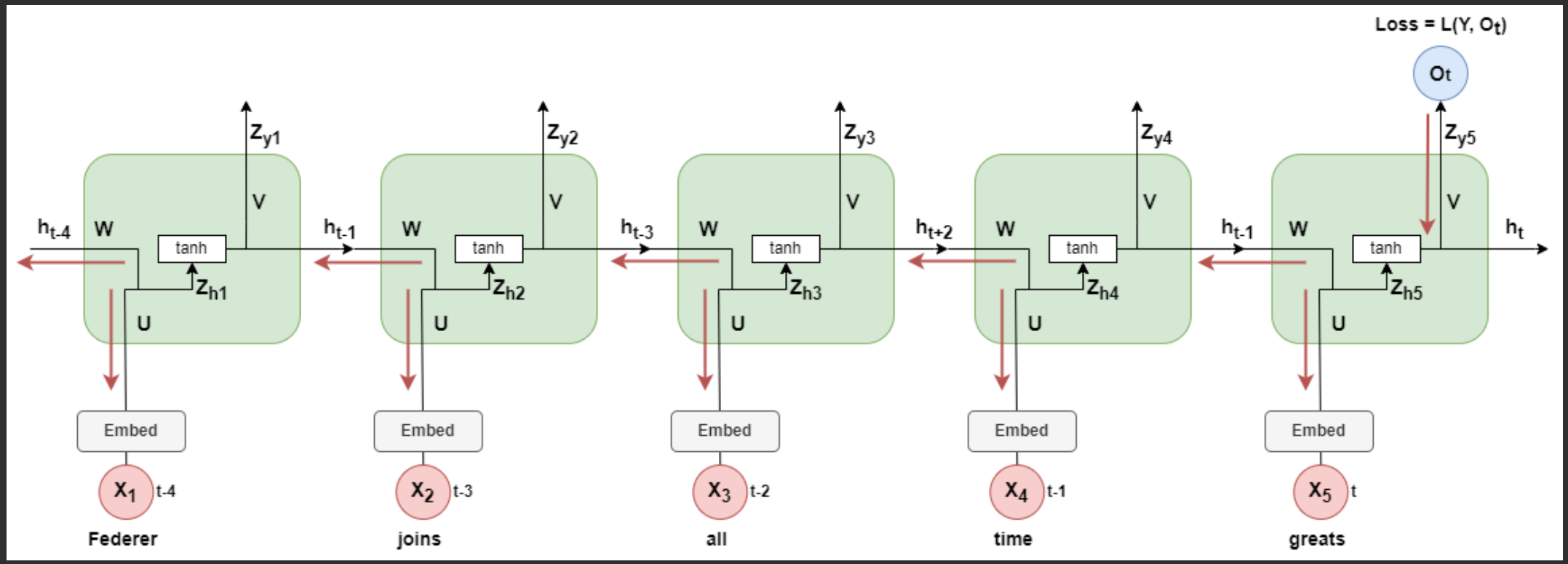
$$O_t = a(V \cdot h_t) + b_2$$

a → tanh / sigmoid

$$h_{t-1} = h_t$$

x_{t_0} = cat

Back propagation through Time (BPTT)



W, U & V

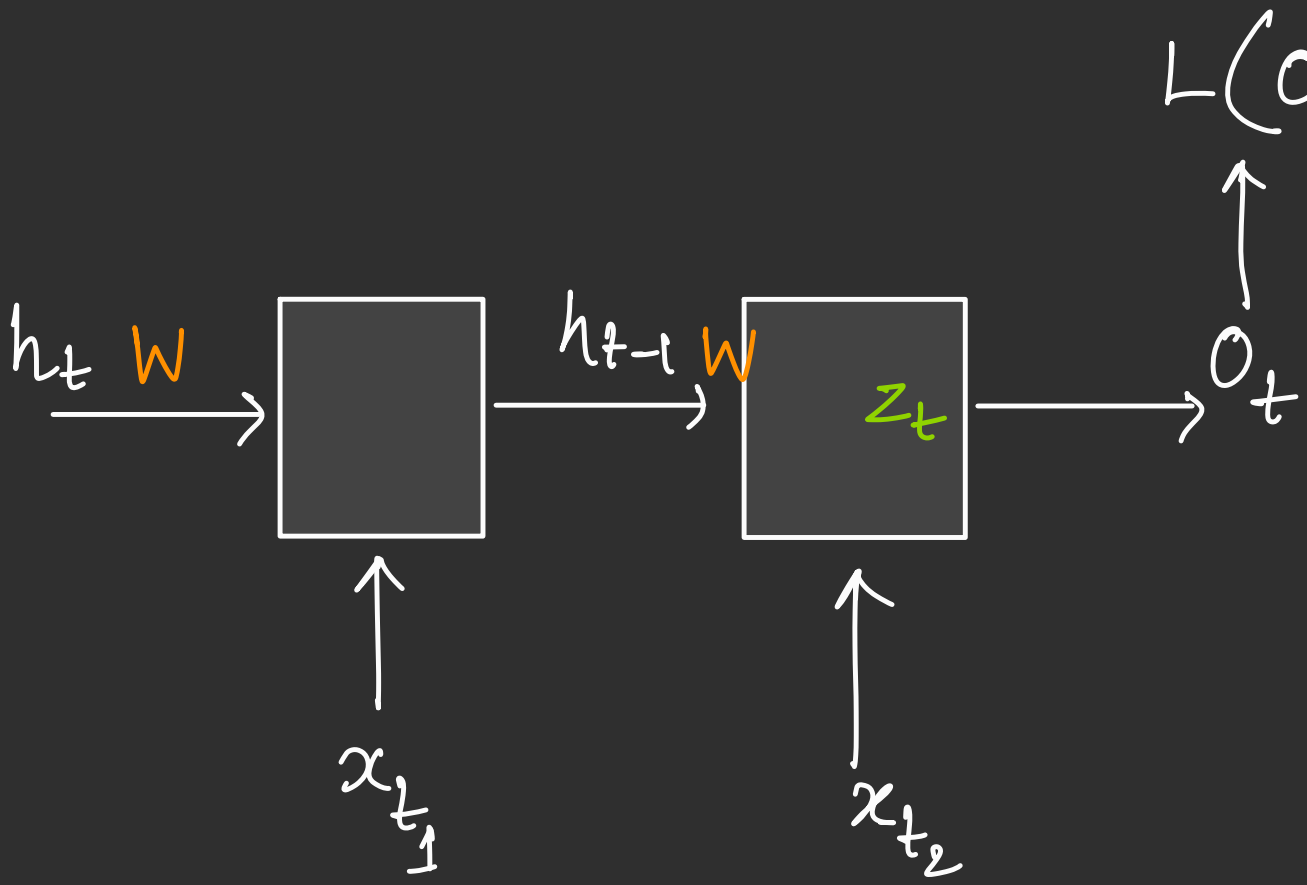
"Gradient descent"

$$\begin{cases} V_{new} = V_{old} - \eta \dfrac{\partial L}{\partial V_{old}} \\ W_{new} = W_{old} - \eta \dfrac{\partial L}{\partial W_{old}} \\ U_{new} = U_{old} - \eta \dfrac{\partial L}{\partial U_{old}} \end{cases}$$

chain rule calculation

We know

Forward pass



$L \rightarrow o_t \rightarrow z_t \rightarrow W$

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial o_t} \times \frac{\partial o_t}{\partial z_t} \times \frac{\partial z_t}{\partial h_t} \times \frac{\partial h_t}{\partial W}$$

$\frac{\partial o_t}{\partial z_t}$: 0.1

$\frac{\partial z_t}{\partial h_t}$: 0.1

$0.1 \times 0.1 \longrightarrow 0.01$ smaller

"vanishing gradient"

It will forget what it has learnt over the course of time

"Longer Sentence problem"

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$

η: 0.1 or 0.01

$w_{new} \approx w_{old}$

"No learning"

Classification

(spam Vs Ham)

final output

sentiment Analysis

Multi-class

$y_1, y_2 \cdots y_n$

(final output)

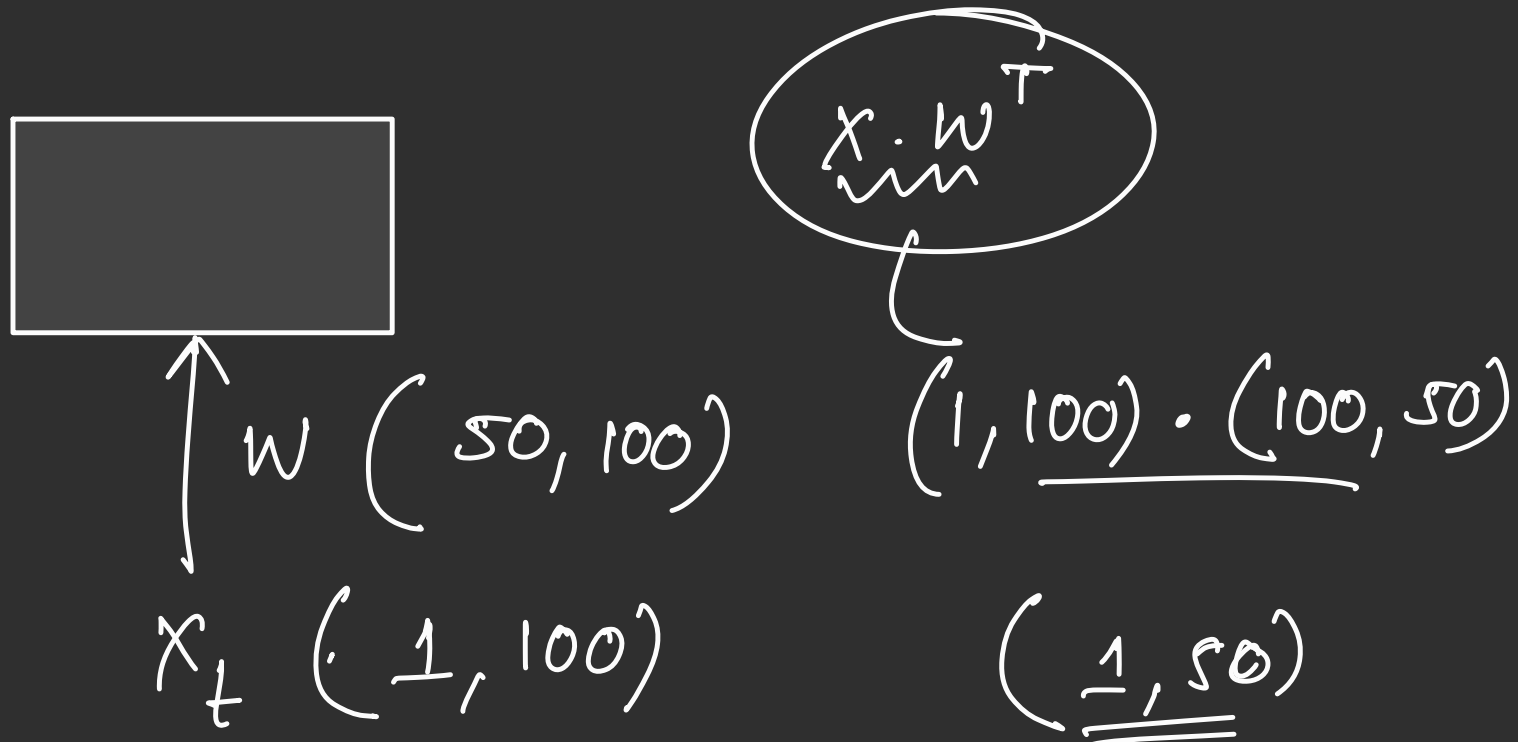
classify the whole sentence.

POS tagging

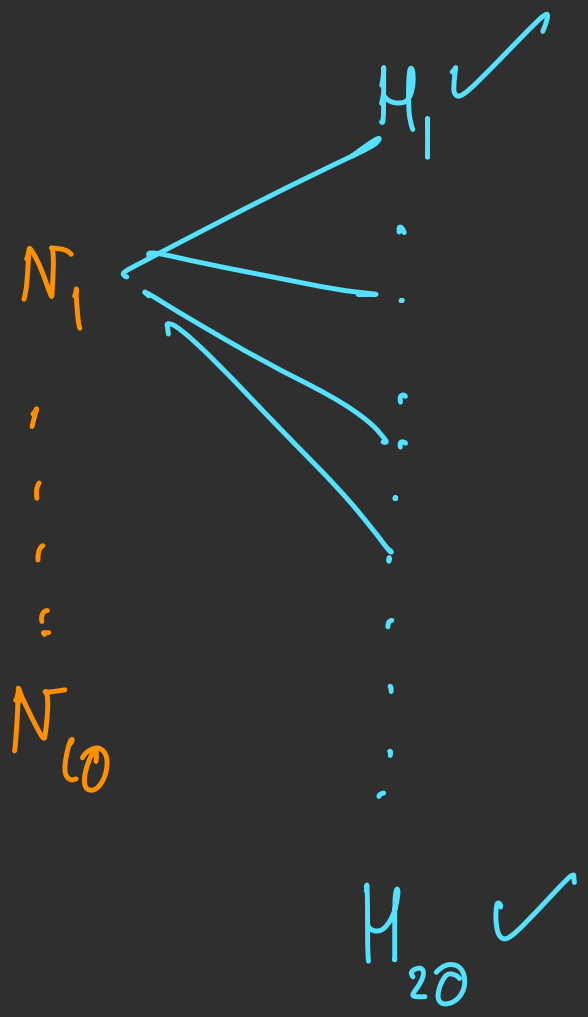
Each word what is it prop, verb

(Multi-class)

$y_1\ y_2 \cdots y_n$ are required


$x \cdot w^T$
$W\ (50, 100)$
$(1, 100) \cdot (100, 50)$
$x_t\ (1, 100)$
$(1, 50)$

feature → 10 fe



SWE

↳ Writing code + debug

↳ Super fast AI

Exp ⟶ prompt

"Saving Time"

↳ utilizing the delta

domain / department / Research

"problem solver"